Project Title: “A Web-Based Construction Marketplace and Project Management System with Client Inquiry, Quotation, and Workforce Recruitment for H&S Construction Services”

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## I. Title of the Study

A Web-Based Construction Marketplace and Project Management System with Client Inquiry, Quotation, and Workforce Recruitment for H&S Construction Services

The title reflects the main functions of the system (client inquiry, quotation, project management, and workforce recruitment) and the client it is intended for.

## II. Introduction

Construction businesses must coordinate client requests, project requirements, estimates, schedules, workers, materials, expenses, and progress updates. When these activities are recorded in separate notebooks, logbooks, receipts, and messaging conversations, information can be difficult to consolidate and retrieve. This makes it challenging to review a project's current status, calculate its accumulated expenses, and give clients timely updates.

The initial interview responses provided by H&S Construction Services describe a largely manual process. The company records client information, schedules, quotations, and expenses in notebooks or logbooks. Material and labor quotations are calculated manually by listing the required materials and quantities, multiplying quantities by unit prices, estimating labor according to the type and duration of work, and adding other possible expenses. The company also keeps purchase receipts, but missing receipts or delayed entries can make it difficult to determine the exact total project cost.

Monitoring projects can be difficult when project sites are far away or when several projects are underway at the same time. H&S Construction Services relies on trusted people at the site and uses Messenger video calls, pictures, and reports to receive and share progress updates. The interview responses also identify possible delays related to material prices, finding materials that meet quality and budget requirements, and having enough workers available when projects overlap.

To address these needs, this study proposes a web-based construction marketplace and project management system for H&S Construction Services. The proposed platform will provide a public-facing portfolio of completed projects, a client inquiry function, quotation preparation and tracking, project records, expense and material monitoring, project progress updates, and workforce recruitment features. The system is intended to centralize relevant information and make it easier for the company and its clients to access project details through a web browser.

The proposal is informed by research on construction information systems and digital project management. Studies have discussed the value of centralized project information, progress tracking, cost and schedule management, and communication tools. However, the proposed system's actual benefits must be evaluated through development and user testing; they should not be assumed before implementation.

## III. Statement of the Problem

H&S Construction Services currently uses notebooks, logbooks, receipts, manual calculations, and online messaging to manage important construction activities. Based on the interview responses supplied for this concept paper, the study will address the following problems:

1. Project details, including client information, schedules, quotations, and expenses, are recorded manually, making information organization and retrieval more difficult.
2. Quotations for materials, labor, and other possible expenses are computed manually, requiring the company to list quantities, unit prices, labor estimates, and additional costs.
3. Expense totals can be difficult to verify when receipts are misplaced or expenses are not recorded promptly.
4. Project progress and worker activities can be challenging to monitor when sites are distant or multiple projects are active.
5. Client updates depend on site visits, messaging, pictures, video calls, and reports rather than a centralized project-status view.
6. The company wants a website where prospective clients can view examples of completed commercial and other construction projects.
7. Workforce availability can become a concern when several projects need workers at the same time.

Therefore, the main research problem is: How can a web-based construction marketplace and project management system help H&S Construction Services organize client inquiries, quotations, project records, expenses, progress updates, and workforce recruitment?

## IV. Objectives of the Study

### General Objective

To design, develop, and evaluate a web-based construction marketplace and project management system with client inquiry, quotation, and workforce recruitment features for H&S Construction Services.

### Specific Objectives

1. To gather and document the business and system requirements of H&S Construction Services through interviews and a review of its current procedures.
2. To develop a public project portfolio where visitors can view selected completed projects and submit inquiries about prospective construction work.
3. To develop a quotation module that records project requirements, material items, quantities, unit prices, estimated labor, other charges, and the resulting estimated total for review by authorized company personnel.
4. To centralize client records, project schedules, project status, material records, receipts or expense entries, and progress updates.
5. To provide a project progress and client-update feature that can present authorized status information, photos, and reports where available.
6. To develop a workforce recruitment feature where applicants can submit their information and the company can review applications and record recruitment status.
7. To implement role-based access so that clients, applicants, and authorized company personnel can access only the functions and information appropriate to their roles.
8. To evaluate the system's functionality, usability, and suitability through testing and feedback from intended users.

## V. Significance of the Study

**H&S Construction Services** – The system may help the company maintain more organized project records, calculate and review recorded expenses, prepare quotations, track project status, and present completed work to prospective clients.

**Clients** – Clients may be able to submit project inquiries online and, when access is provided by the company, view relevant quotation and progress information without relying solely on in-person visits or separate messages.

**Workers and applicants** – The workforce recruitment feature may provide a single channel for submitting applications and allow the company to review applicant information when workers are needed.

**Project personnel** – Authorized personnel may use centralized project and material records to support coordination, review costs, and communicate updates.

**Future researchers** – The study may serve as a reference for future work involving web-based construction management, small-business digitization, quotation workflows, and workforce recruitment.

## VI. Scope and Delimitations

### Scope

The proposed system will be a web-based platform designed for H&S Construction Services. Subject to confirmation during requirements gathering, its main modules will include:

- **Public construction portfolio** – displays approved photos, descriptions, and categories of completed projects.
- **Client inquiry** – allows visitors to submit contact details and basic project requirements for company follow-up.
- **Quotation management** – records material descriptions, quantities, unit prices, labor estimates, other expenses, and estimated totals; allows authorized personnel to review and update quotations and record their status.
- **Project management** – stores client and project information, planned schedules, status, assigned personnel, and project notes.
- **Expense and material tracking** – records material purchases, quantities, costs, dates, and receipt references or uploaded receipt images where supported.
- **Progress updates** – records project updates and, where available, attaches photos or reports that authorized users can view.
- **Workforce recruitment** – accepts worker applications and allows authorized personnel to review applicant details and recruitment status.
- **User and access management** – provides appropriate access for administrators or company personnel, clients, and applicants, based on the final requirements.

**Status of requirements.** The needs reported in the H&S Construction Services interview (manual recordkeeping, manual quotation and expense computation, remote progress monitoring through messaging, worker availability concerns, and a website showing completed projects) are treated as the basis of this study. Modules and features beyond these reported needs are proposals by the researchers and remain subject to the client's approval during requirements gathering.

**Delimitations**

- The system will support recordkeeping and estimation; it will not guarantee that a quotation reflects final supplier prices, unforeseen site conditions, or the final cost of construction.
- The system will not automatically verify the quality or market price of materials. Company personnel will remain responsible for checking material specifications and prices.
- Progress information will depend on updates entered or uploaded by authorized users. The system cannot independently confirm the physical condition of a remote construction site.
- The initial version will not include automatic payroll, full accounting, tax filing, online payment processing, computer-aided design, or building information modeling unless these are added after the scope is approved.
- Workforce recruitment will organize applications and their status; it will not automatically certify applicants' skills, qualifications, or availability.
- The system's final technology stack, hosting environment, notifications, privacy controls, and user roles will be confirmed during system analysis and design.

## VII. Methodology

This study will adopt a Developmental Research approach using the Agile Development Methodology to facilitate continuous development, testing, and feedback from H&S Construction Services.

**Development Tools**

- **Frontend:** HTML and JavaScript
- **Backend:** PHP with Bootstrap
- **Database:** MySQL

### Research Process

1. **Requirements Gathering** – Interviews with the H&S Construction Services owner and employees.
2. **System Design** – Use case diagrams, database schema, and system architecture.
3. **Development** – Incremental feature implementation using Agile sprints.
4. **Testing** – Unit testing, load testing, stress testing, security testing, and user acceptance testing.
5. **Evaluation** – Surveys and feedback from pilot users to measure usability and effectiveness.

## VIII. Review of Related Literature

### Construction Project Information and Digital Tools

Construction projects generate information about clients, schedules, materials, costs, progress, and documents. A study of information technology systems in multi-sector construction projects describes the role of digital tools in progress tracking, status reporting, communication, information exchange, and maintaining a centralized repository of project documentation. These functions are relevant to the proposed system because H&S Construction Services currently spreads records across notebooks, logbooks, receipts, and messaging conversations (Waszkiewicz & Gumienny, 2021).

### Document Management in Small Construction Companies

Research on document management in small construction companies examined paper-based and electronic approaches to storing, controlling, coordinating, processing, and retrieving project information. Its focus is closely related to the client's reported difficulty in finding exact expense totals when receipts are lost or entries are delayed. A centralized digital record may make it easier to retrieve project information, although the result will depend on consistent data entry and adoption by users (Ahmad et al., 2017).

### Information and Communication Technology Adoption by Small Builders

Rajendra et al. (2022) investigated technology readiness and intention to use information and communication technologies among small construction businesses. Their study used an online survey of 110 Australian builders and six expert interviews. The authors reported that readiness dimensions such as optimism, innovativeness, and insecurity were important determinants of intention to use information and communication technologies. This suggests that a system for

a small construction company should be practical and understandable, and that implementation should consider user concerns and available resources rather than assuming that introducing software alone will ensure adoption.

### Project Management Information Systems and Cost/Schedule Tracking

Research on project management information systems for industrial construction discusses functions that connect time management, cost management, project status, resources, and project documentation. Such functions are relevant to the proposed project-record, quotation, expense, and progress modules. For H&S Construction Services, the first version should focus on the company's stated needs and avoid adding complex functions that its current workflow does not require (Choi & Ha, 2021).

### Construction Information Systems and Operational Efficiency

Vaughan et al. (2013) discussed construction information management systems and reported, in their case study, that using such a system with mobile technologies increased efficiency and reduced clerical time for operations-level personnel. This supports investigating whether centralizing records and making updates accessible through a web-based system can reduce some manual administrative work. The results of that separate case study do not establish that the proposed H&S Construction Services system will achieve the same outcomes; these must be assessed during its own evaluation.

### Synthesis and Research Gap

The literature reviewed supports the relevance of organized digital records, progress reporting, communication, and project cost and schedule information in construction management. The client interview identifies a specific local workflow in which quotations and expenses are computed manually, receipts are retained as evidence, remote progress is communicated through people at the site and messaging tools, and the company wants a public portfolio of completed projects. The proposed system brings these requirements together in one web-based platform and adds client inquiry and workforce recruitment functions. The available literature does not prove that an identical system has already solved the particular problems of H&S Construction Services; therefore, the study will design and evaluate a solution based on the client's validated requirements.

## IX. Expected Output

The expected output is a functional web-based construction marketplace and project management system for H&S Construction Services. Subject to final requirements approval, deliverables will include:

- A public website presenting selected completed construction projects.
- A client inquiry form and an inquiry-management view for authorized company personnel.
- A quotation module for recording materials, quantities, unit prices, estimated labor, additional costs, and quotation status.
- A project management module for client details, schedules, assignments, status, and notes.
- An expense and material record module with receipt references or supported receipt uploads.
- A project progress update area for authorized updates, photos, and reports.
- A workforce recruitment module for applicant submissions and recruitment-status tracking.
- User access controls, test documentation, and basic user instructions.

These are intended deliverables. The system's actual performance, usability, and operational impact will be stated only after implementation and evaluation.

## X. References

Ahmad, H. S., Bazlamit, I. M., & Ayoush, M. D. (2017). Investigation of document management systems in small size construction companies in Jordan. *Procedia Engineering, 182*, 3–9. https://doi.org/10.1016/j.proeng.2017.03.101

Brooke, J. (1996). SUS: A quick and dirty usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability evaluation in industry* (pp. 189–194). Taylor & Francis.

Choi, J., & Ha, M. (2021). Validation of project management information systems for industrial construction projects. *Journal of Asian Architecture and Building Engineering, 21*(5), 2046–2057. https://doi.org/10.1080/13467581.2021.1941999

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Vaughan, J. L., Leming, M. L., Liu, M., & Jaselskis, E. J. (2013). Cost-benefit analysis of construction information management system implementation: Case study. *Journal of Construction Engineering and Management, 139*(4). https://ascelibrary.org/doi/10.1061/%28ASCE%29CO.1943-7862.0000611

Waszkiewicz, M., & Gumienny, K. (2021). Benefits of using IT systems in multi-sector projects: A case study of the Polish construction industry. *Procedia Computer Science, 192*, 4269–4278. https://doi.org/10.1016/j.procs.2021.09.203

## XI. Proposed Workflows and Conceptual Framework

### A. Proposed Workflows

Figure 1 shows the proposed workflow of the system. A visitor can view the public portfolio and submit an inquiry. Authorized company personnel review the inquiry, prepare a quotation, and record its status. Once a quotation is accepted, a project record is created, and expenses, material purchases, and progress updates are recorded and made available to authorized users. Applicants may separately submit worker applications, which authorized personnel review when workers are needed. The workflow is a proposal and remains subject to the approval of H&S Construction Services.

*Figure 1. Proposed workflow of the system*

### B. Conceptual Framework

Figure 2 presents the conceptual framework of the study using the input–process–output model. The inputs are the interview responses from H&S Construction Services, client inquiries and project requirements, material and labor information and receipts, project updates, and worker applications. These pass through the research process described in the Methodology: requirements gathering, system design, development, testing, and evaluation. The expected outputs are the web-based system and its organized records, centralized progress updates, and recruitment tracking.

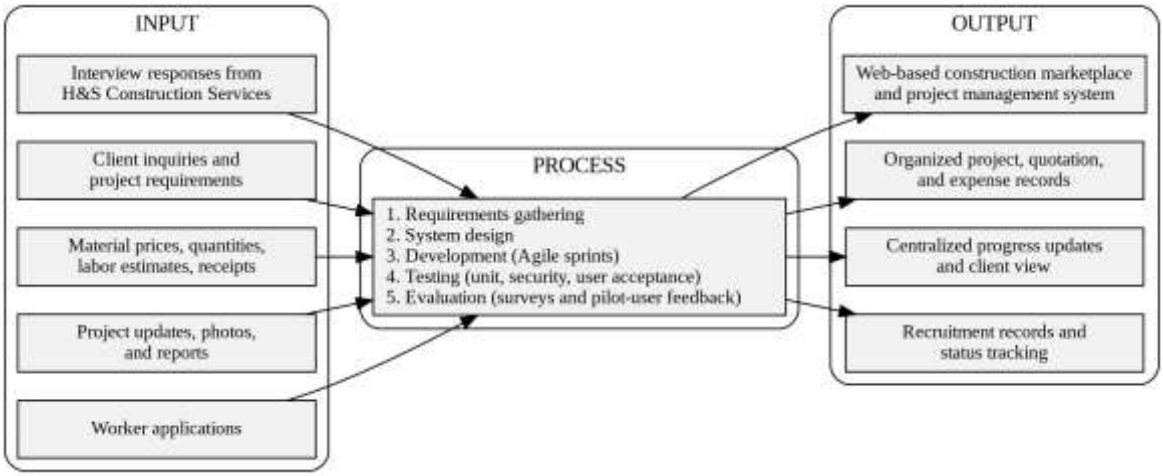



*Figure 2. Conceptual framework of the study*

## XII. Appendices

### Appendix A – Consolidated Data

Summary of the interview findings from H&S Construction Services used in this study:

- **Current process:** Client information, schedules, quotations, and expenses are recorded in notebooks or logbooks. Quotations are computed manually by listing materials and quantities, multiplying by unit prices, estimating labor by type and duration of work, and adding other possible expenses. Purchase receipts are kept as records.
- **Challenges:** Missing receipts or delayed entries make exact project costs hard to determine. Monitoring distant sites or several simultaneous projects is difficult. Possible delays come from material prices, finding materials that meet quality and budget requirements, and worker availability when projects overlap.
- **Current progress reporting:** The company relies on trusted people at the site and uses Messenger video calls, pictures, and reports.
- **Stated need:** A website where prospective clients can view examples of completed projects.

## Appendix B – Use Case Diagram

The use case diagram shows how each user role interacts with the proposed system features, ensuring role-based access.

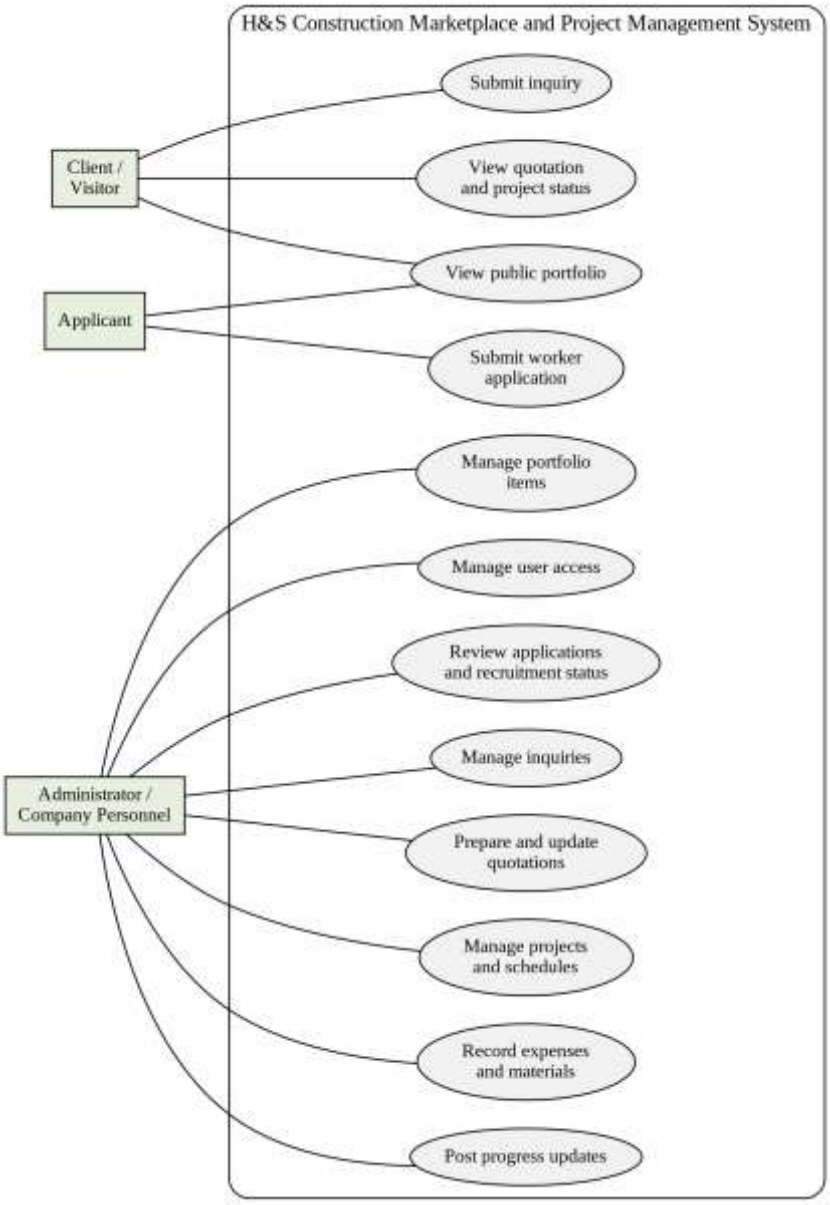



*Figure 3. Proposed use case diagram*

## Appendix C – Entity-Relationship Diagram (ERD)

The ERD illustrates the proposed database tables and how project data flows from inquiry and quotation to expenses and progress updates. The final tables will be confirmed during system design.

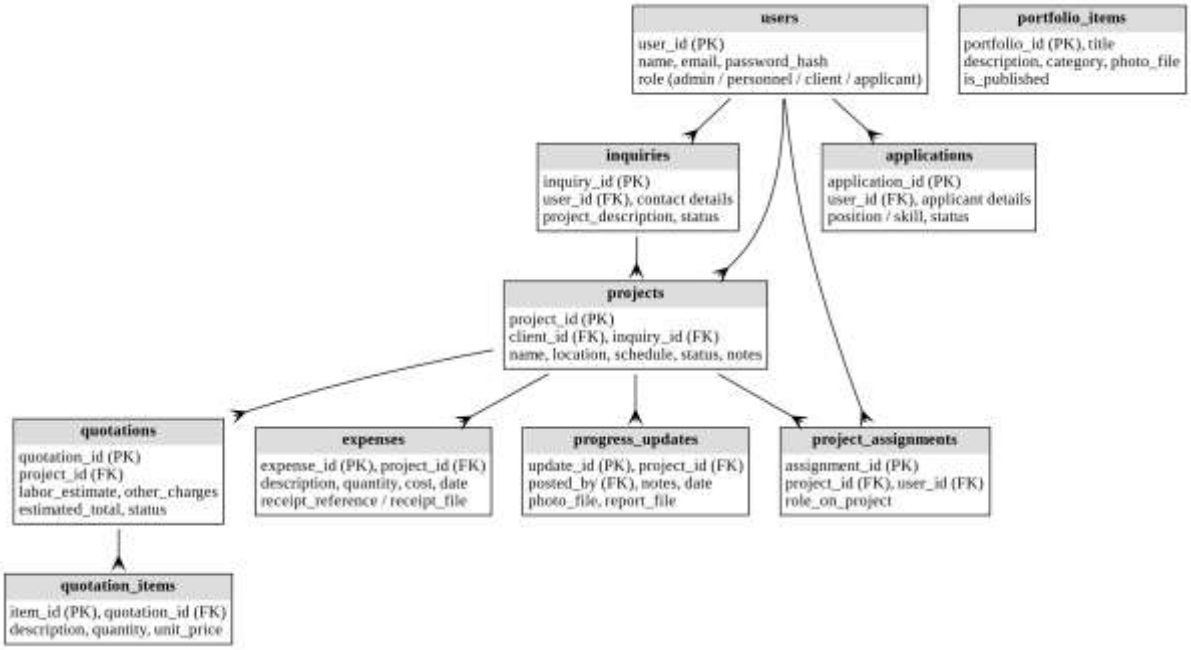



*Figure 4. Proposed entity-relationship diagram*

## Appendix D – System Architecture

The proposed system follows a three-tier architecture using the tools stated in the Methodology: HTML, JavaScript, and Bootstrap for the frontend; PHP for the backend; and MySQL for the database.

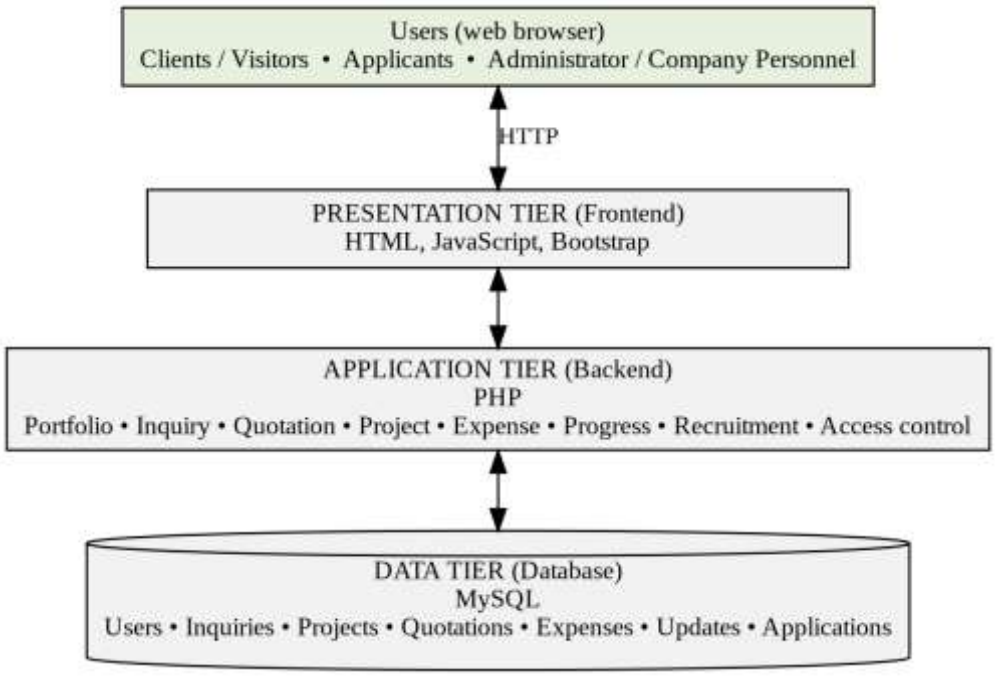



*Figure 5. Proposed three-tier system architecture*

## Appendix E – Figma UI/UX Prototypes

The figures below are low-fidelity wireframes of the planned key screens. They show the intended layout and content of each screen. The final Figma designs may differ in colors, spacing, and details and will be confirmed with H&S Construction Services.

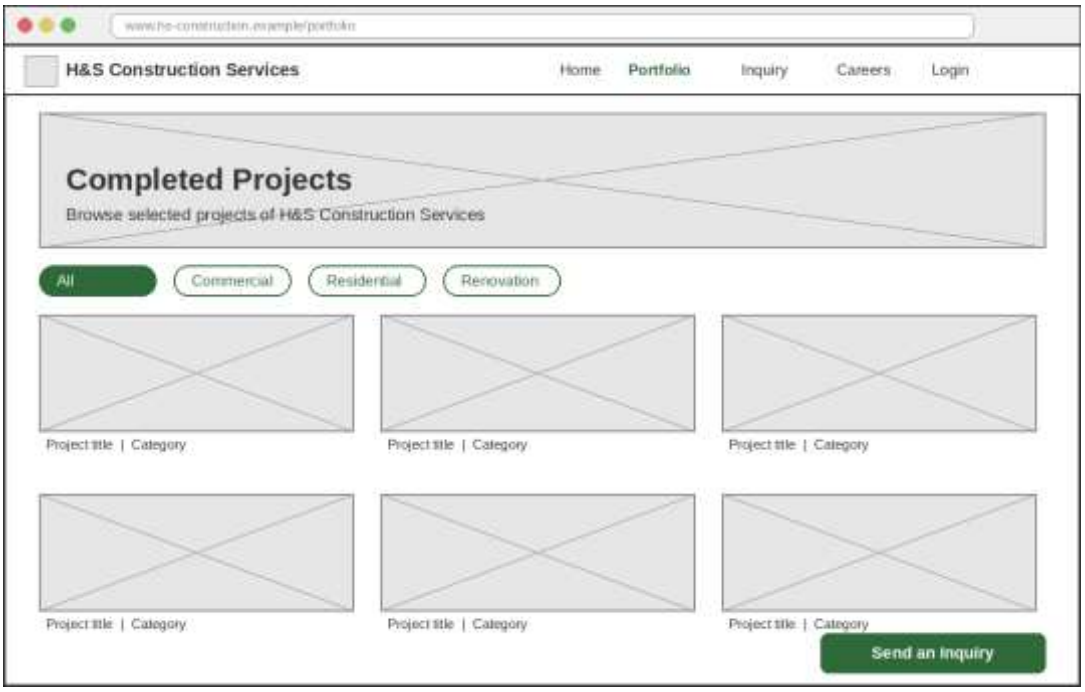



*Figure 6. Public project portfolio page. This wireframe shows the planned page where visitors browse completed projects.*

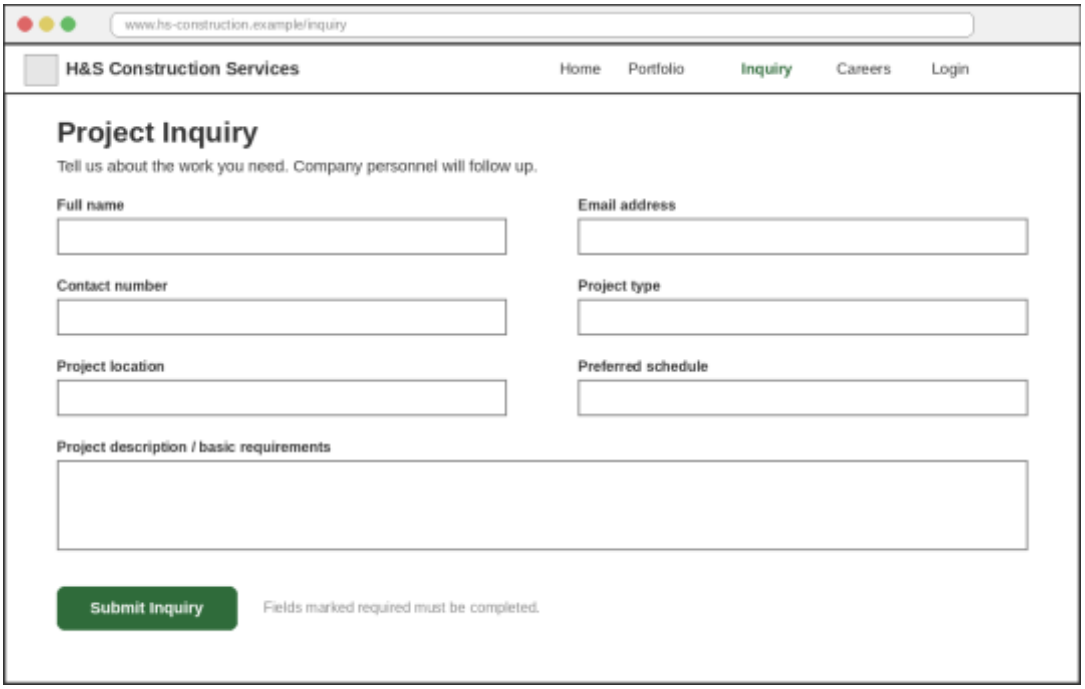



*Figure 7. Client inquiry form. This wireframe shows the planned form for submitting contact details and basic project requirements.*

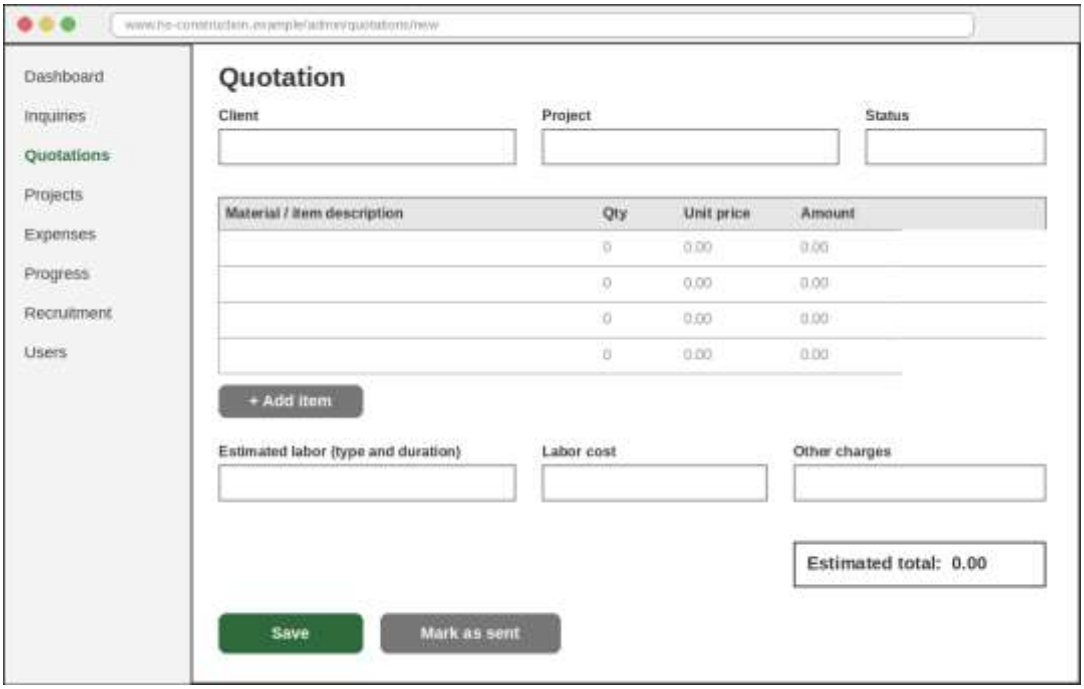



*Figure 8. Quotation management screen. This wireframe shows the planned entry of materials, quantities, unit prices, labor, other charges, and quotation status.*

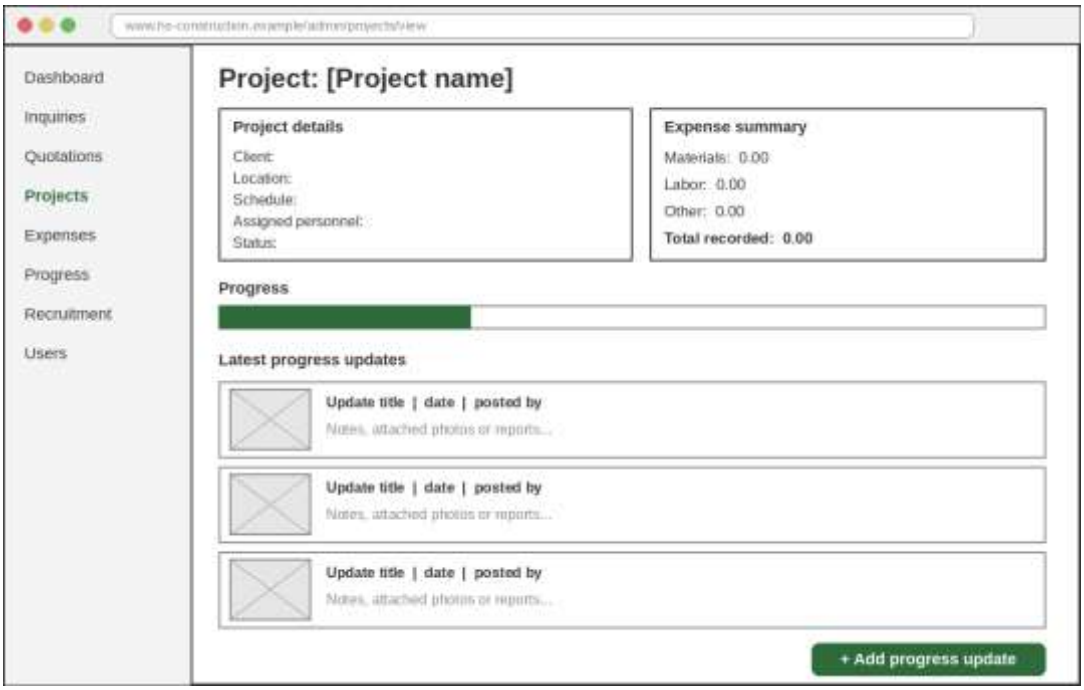



*Figure 9. Project dashboard. This wireframe shows the planned view of project details, recorded expenses, progress, and updates.*

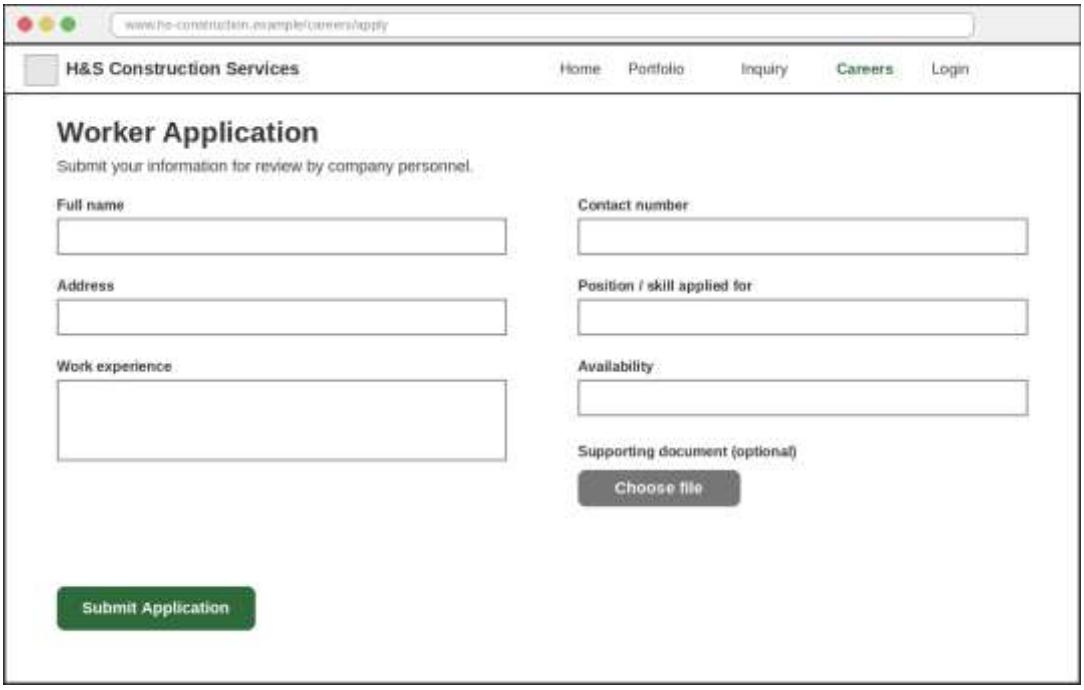



*Figure 10. Worker application form. This wireframe shows the planned form for submitting a worker application.*

## Appendix F – Testing Instruments

The instruments below will be used in the testing and evaluation phases stated in the Methodology.

### F.1 System Usability Scale (SUS) Questionnaire

The System Usability Scale (Brooke, 1996) will be answered by pilot users after using the system. Instruction: Rate each statement from 1 (strongly disagree) to 5 (strongly agree).

| No. | Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| 1 | I think that I would like to use this system frequently. | | | | | |
| 2 | I found the system unnecessarily complex. | | | | | |
| 3 | I thought the system was easy to use. | | | | | |
| 4 | I think that I would need the support of a technical person to be able to use this system. | | | | | |
| 5 | I found the various functions in this system were well integrated. | | | | | |
| 6 | I thought there was too much inconsistency in this system. | | | | | |
| 7 | I would imagine that most people would learn to use this system very quickly. | | | | | |
| 8 | I found the system very cumbersome to use. | | | | | |
| 9 | I felt very confident using the system. | | | | | |
| 10 | I needed to learn a lot of things before I could get going with this system. | | | | | |

### F.2 Sample Test Cases

The test cases below are samples for validating the main modules. The Result column will be completed during testing.

| Test ID | Module | Test case | Expected result | Result |
|---|---|---|---|---|
| TC-01 | Portfolio | Open the public portfolio page as a visitor. | Approved completed projects are displayed with photos, descriptions, and categories. | |
| TC-02 | Inquiry | Submit the inquiry form with all required fields completed. | The inquiry is saved and appears in the inquiry list of authorized personnel. | |
| TC-03 | Inquiry | Submit the inquiry form with a required field left blank. | The system shows a validation message and does not save the inquiry. | |

| Test ID | Module | Test case | Expected result | Result |
|---|---|---|---|---|
| TC-04 | Quotation | Enter material items with quantities and unit prices, labor, and other charges. | Item amounts and the estimated total are computed correctly. | |
| TC-05 | Quotation | Change the status of a quotation. | The new status is saved and displayed. | |
| TC-06 | Project | Create a project record with schedule and assigned personnel. | The project is saved and can be viewed in the project list. | |
| TC-07 | Expense | Add a material expense with a receipt reference or file. | The expense is saved and included in the project expense total. | |
| TC-08 | Progress | Post a progress update with a photo or report. | The update is saved and visible to authorized users. | |
| TC-09 | Recruitment | Submit a worker application, then update its status as personnel. | The application is saved and the updated status is displayed. | |
| TC-10 | Access control | A client tries to open a page reserved for company personnel. | Access is denied or the user is redirected. | |
| TC-11 | Security | Log in with an incorrect password. | Login is rejected and an error message is shown. | |